

Protective effect of fruits, vegetables and the Mediterranean diet on asthma and allergies among children in Crete

Background Atopy is not uncommon among children living in rural Crete, but wheeze and rhinitis are rare. A study was undertaken to examine whether this discrepancy could be attributed to a high consumption of fresh fruit and vegetables or adherence to a traditional Mediterranean diet. **Methods** A cross-sectional survey was performed in 690 children aged 7–18 years in rural Crete. Parents completed a questionnaire on their child's respiratory and allergic symptoms and a 58-item food frequency questionnaire. Adherence to a Mediterranean diet was measured using a scale with 12 dietary items. Children underwent skin prick tests with 10 common aeroallergens. **Results** 80% of children ate fresh fruit (and 68% vegetables) at least twice a day. The intake of grapes, oranges, apples, and fresh tomatoes—the main local products in Crete—had no association with atopy but was protective for wheezing and rhinitis. A high consumption of nuts was found to be inversely associated with wheezing (OR 0.46; 95% CI 0.20 to 0.98), whereas margarine increased the risk of both wheeze (OR 2.19; 95% CI 1.01 to 4.82) and allergic rhinitis (OR 2.10; 95% CI 1.31 to 3.37). A high level of adherence to the Mediterranean diet was protective for allergic rhinitis (OR 0.34; 95% CI 0.18 to 0.64) while a more modest protection was observed for wheezing and atopy. **Conclusion** The results of this study suggest a beneficial effect of commonly consumed fruits, vegetables and nuts, and of a high adherence to a traditional Mediterranean diet during childhood on symptoms of asthma and rhinitis. Diet may explain the relative lack of allergic symptoms in this population.